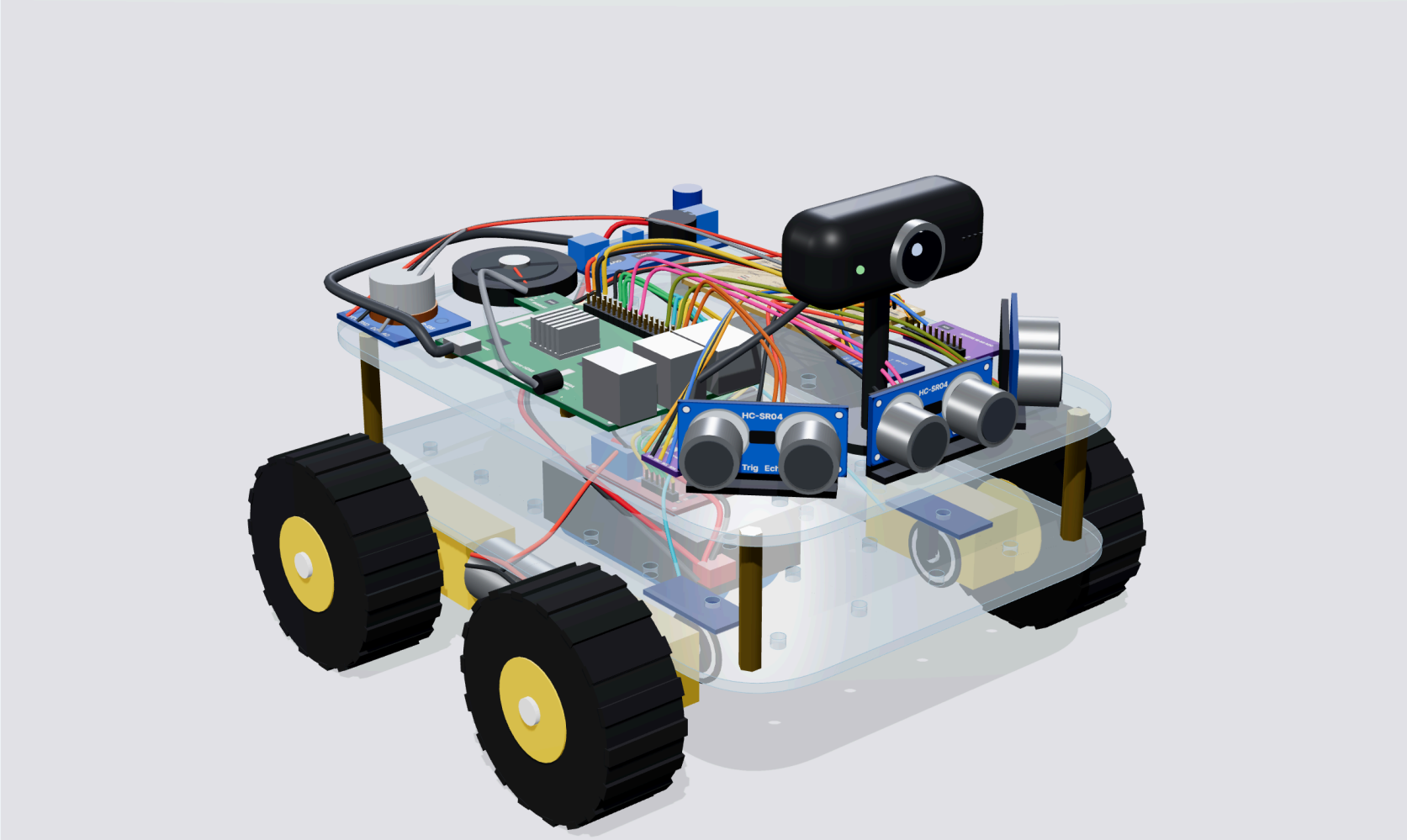
RAQIB · THE FINISHED ROBOT · 3D VIEW

The finished robot

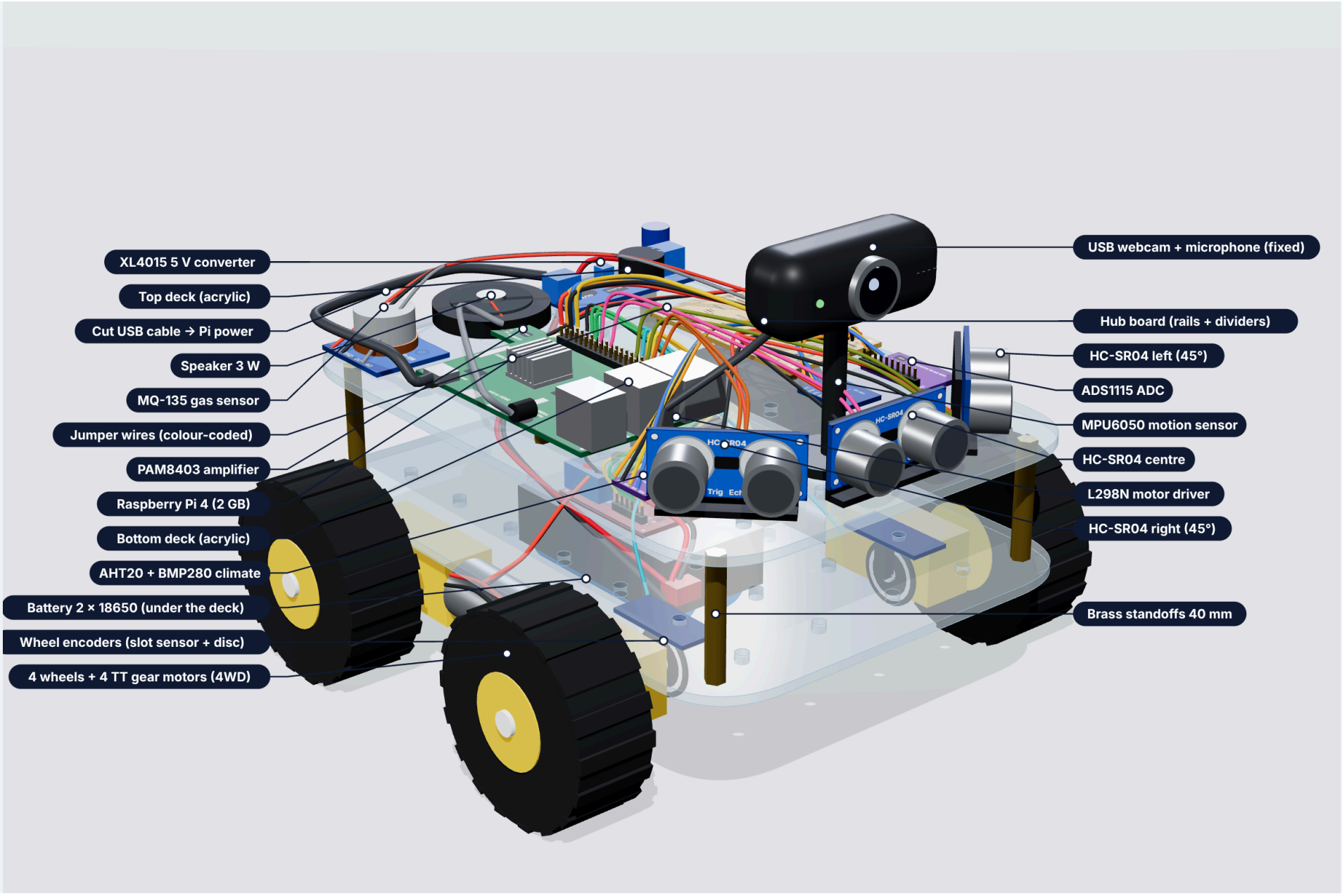
Raqib, the budget AI inspection robot, fully built and wired: four-wheel drive, two acrylic decks, three ultrasonic sensors and a USB webcam at the front, the Raspberry Pi and sensors on top, motors and battery below.



A 3D model at real size (mm), built from the parts and layout of the build guide. Board colours and printed labels follow the common versions of each part; your boards may differ slightly.

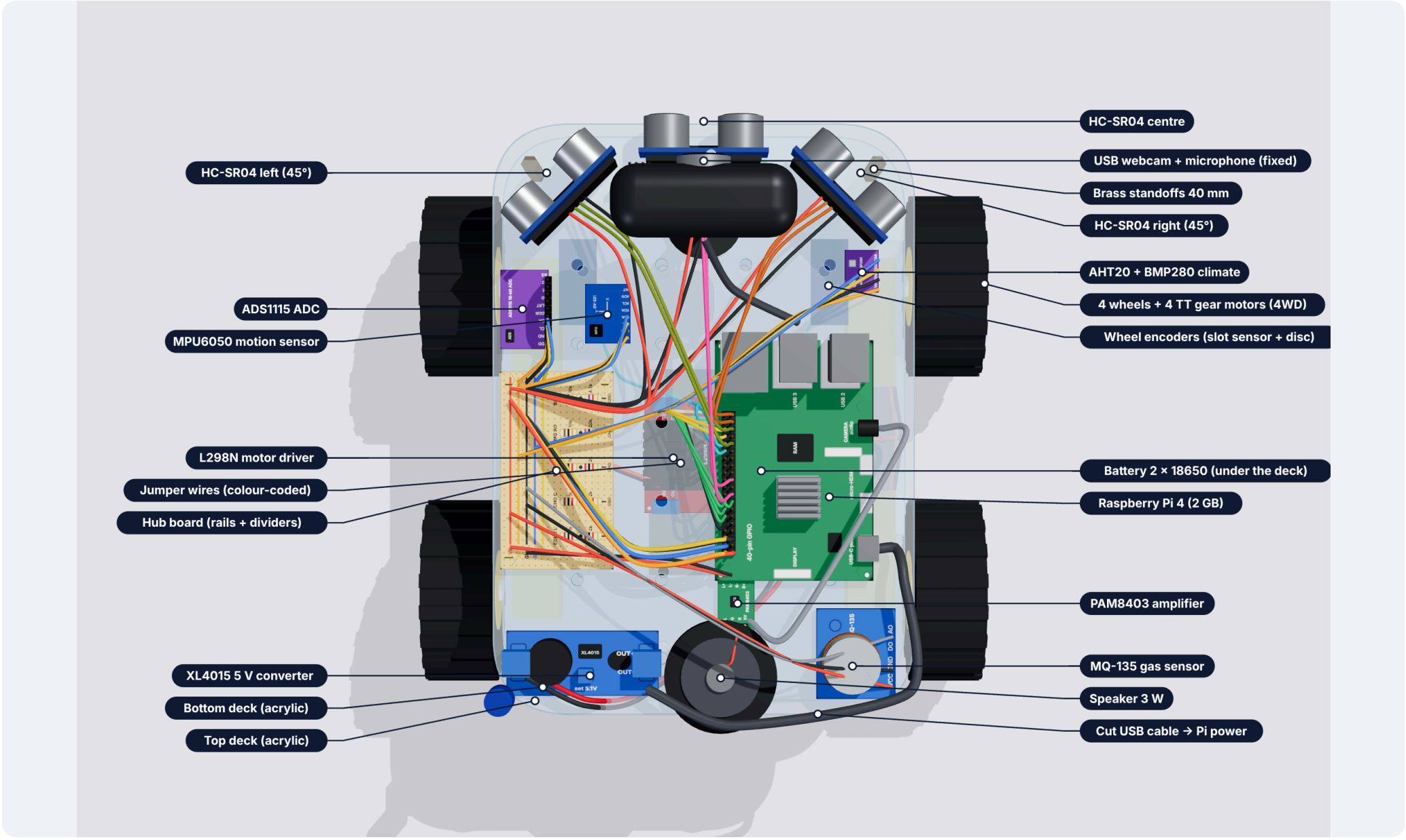
Every part, labelled

All parts in their final place. Front of the robot = the side with the webcam.



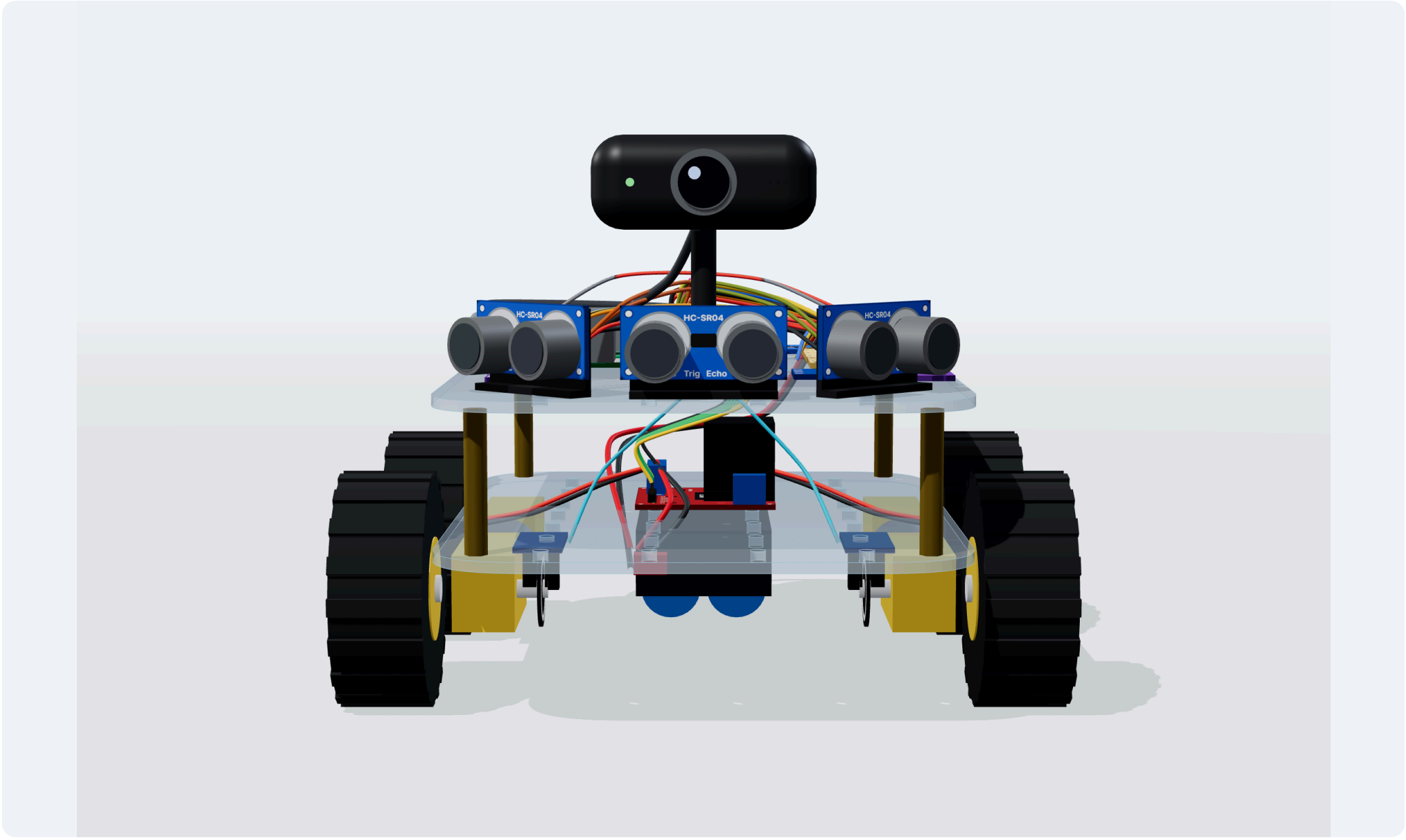
Seen from above: how the top deck is organised

Front (top of the picture): three HC-SR04 fanned left / centre / right and the webcam on its post. Middle: Raspberry Pi with its 40-pin header facing the hub board, MPU6050 near the centre, ADS1115, AHT20. Back: XL4015, speaker + PAM8403, MQ-135 (warm parts away from the AHT20).



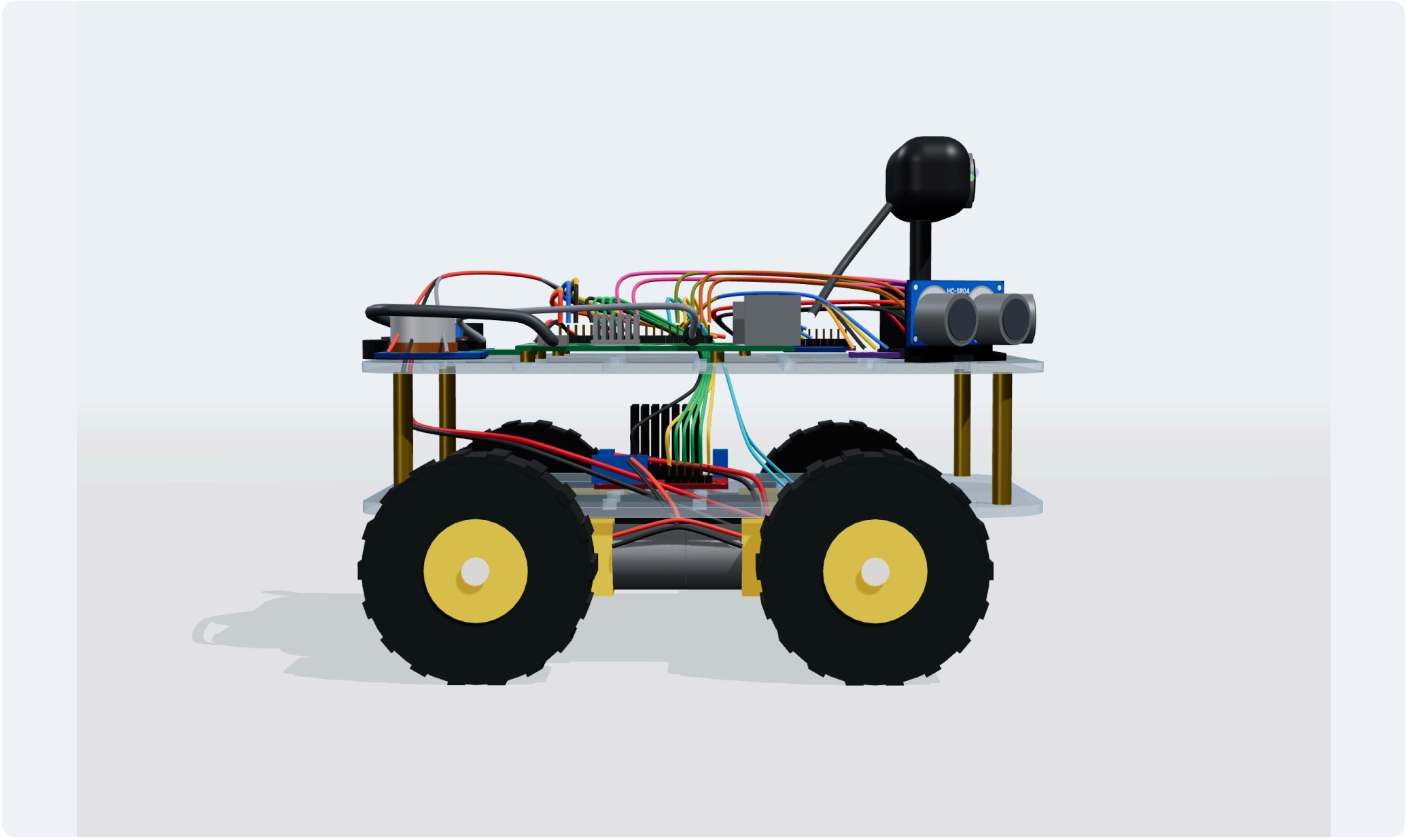
Front view

What the robot looks at: the centre sonar straight ahead, the left and right sonars turned 45°, the webcam above them. The camera is fixed: the robot turns its whole body to look around.



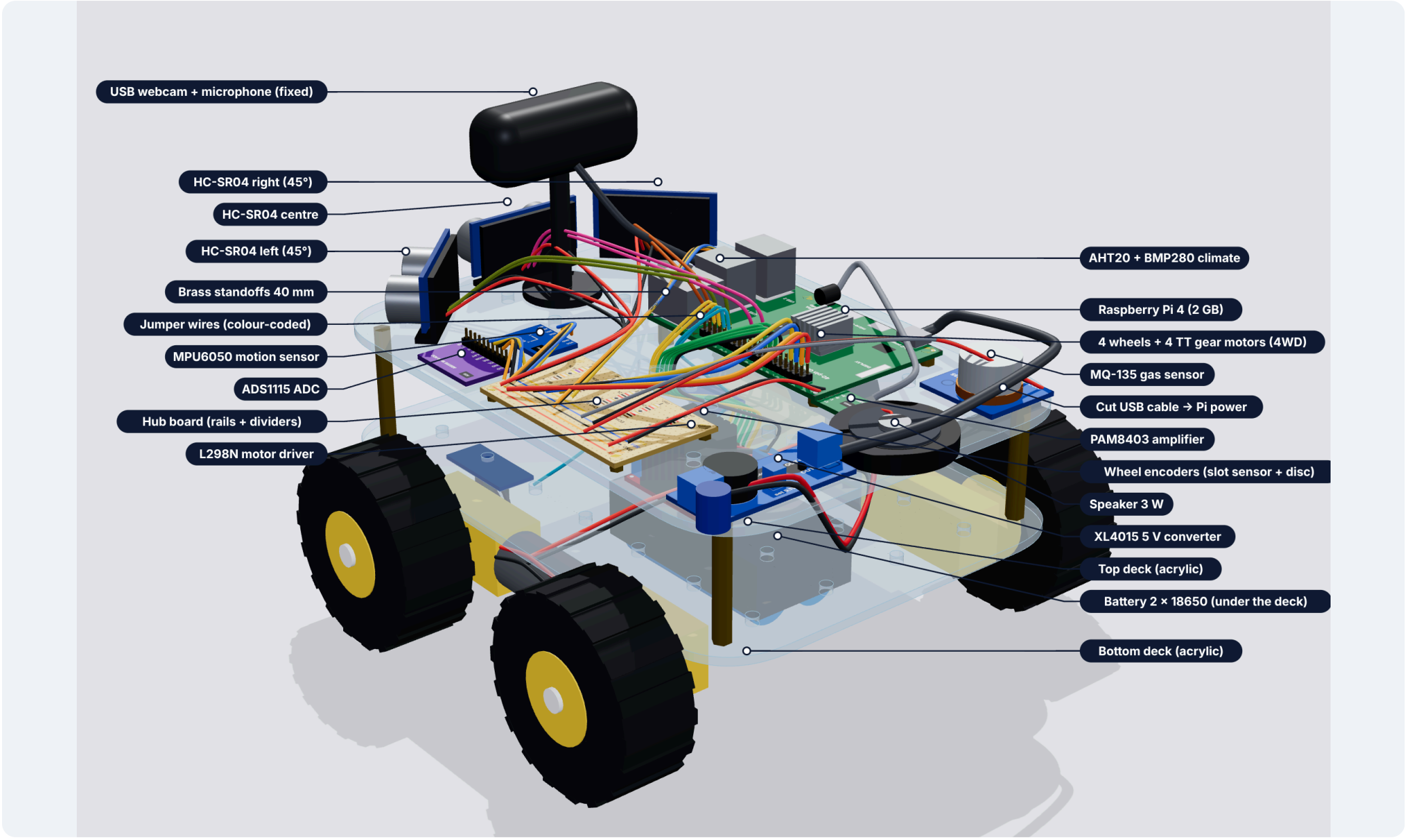
Side view

Two decks on 40 mm brass standoffs. Bottom deck: L298N motor driver; under it the battery and the four TT motors with their wheels (4WD). Top deck: Pi, sensors, speaker, webcam.



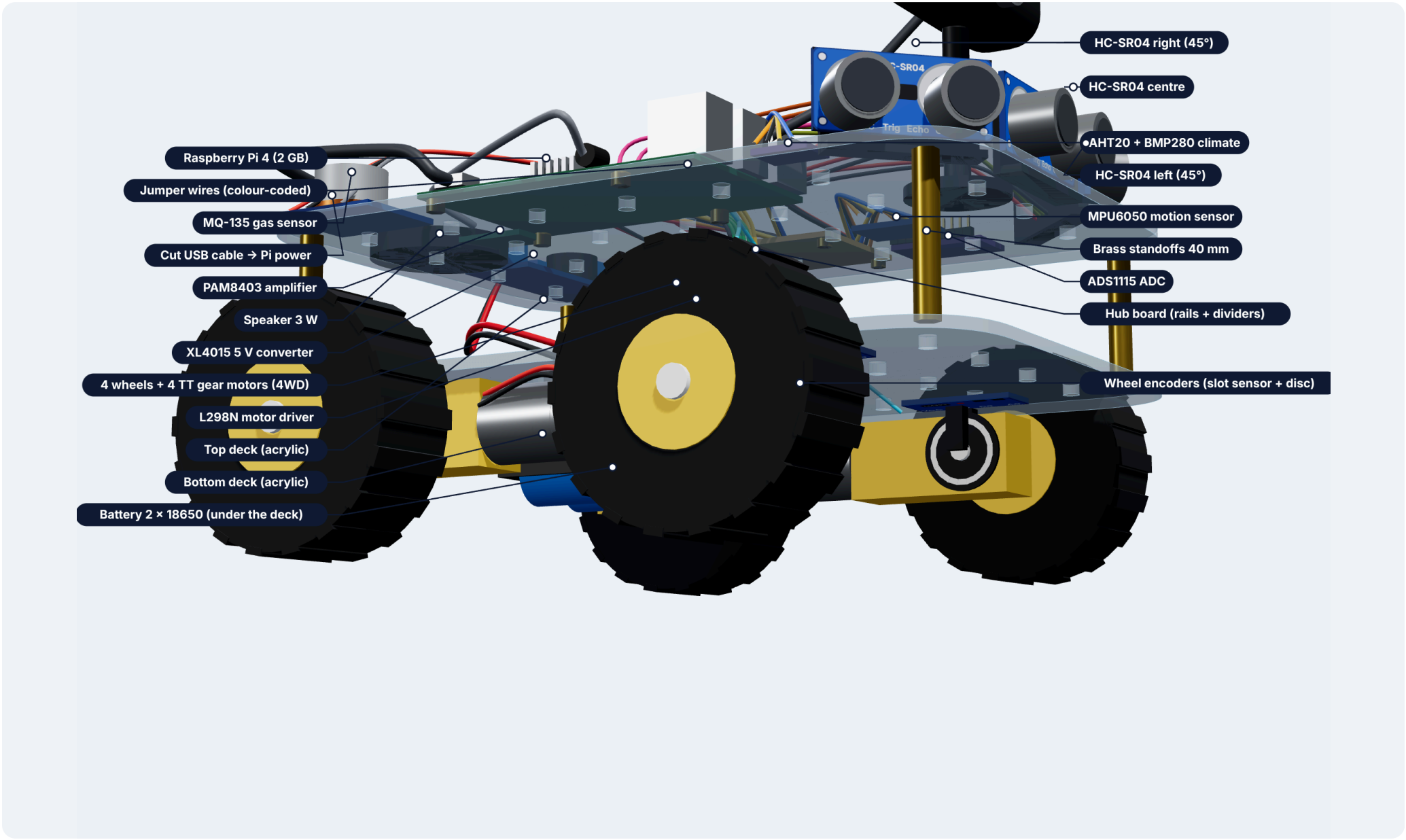
From the back

The XL4015 converter, the cut USB cable to the Pi's power socket, the MQ-135 and the speaker sit at the back; the wires from the Pi go down through the deck to the L298N.



Underneath

The 2 × 18650 holder hangs under the bottom deck with its switch reachable; four TT motors, two per side; the front pair carries the encoder discs on their inner shafts, read by the slot sensors.



Close-up of the top deck

Colour-coded jumper wires from the Pi's 40-pin header: red 5 V, orange 3.3 V, black GND, blue SDA, yellow SCL, green motor direction, amber motor speed, pink / olive / brown sonars, cyan encoders.

